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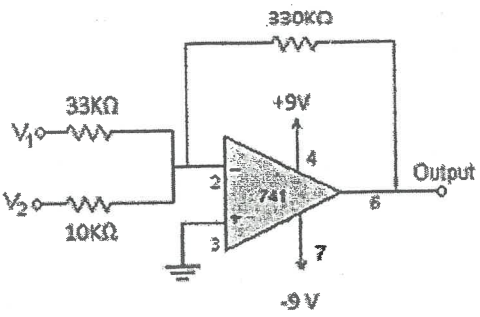
THIRD SEMESTER B. TECH EXAMINATION, DECEMBER-2024

ANALOG AND DIGITAL CIRCUITS-2

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5x2=10**
1.	**a.**	Design LPF, for cut off frequency F_L= 8KHz and gain of 1.58.	L3	CO2	**02**
	b.	Calculate the output voltage if V_1=-0.2 V and V_2=0 V for the circuit shown in Fig 1b.  Fig 1b	L3	CO4	**02**
	c.	Design an amplifier with a voltage gain of -10 and input resistance equal to 10kΩ.	L2	CO1	**02**
	d.	What do you mean by precision rectifier?	L2	CO1	**02**
	e.	Differentiate the concept of feedback with respect to the Op-amp.	L3	CO2	**02**
		PART-B			**6X5=30**
2.	**a.**	Derive an expression for the gain of non-inverting amplifier.	L2	CO1	**05**
		OR			
	b.	Write a short note on the concept of voltage divider using Op-amp.	L2	CO1	**05**
3.	**a.**	Explain the operation Op-amp as an integrator and differentiator along with its output wave forms.	L2	CO1	**05**
		OR			
	b.	Briefly explain the concepts of Operational Transconductance Amplifier (OTA).	L2	CO1	**05**
4.	**a.**	Classify active filters and briefly explain each type .	L3	CO6	**05**
		OR			
	b.	Describe how triangle/square wave is generated using Op-amp.	L3	CO6	**05**
5.	**a.**	Write a short note on applications of 555 timer.	L3	CO6	**05**
		OR			
	b.	Explain A-D converter using Op-amp.	L3	CO6	**05**

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6.	a.	What do you mean by Field Programmable Gate Array (FPGA)? Explain in brief.	L2	CO5	05
		OR			
	b.	Explain the working of Complementary MOS (CMOS) transistor.	L2	CO5	05
7.	a.	Describe the operation and applications of astable multivibrator using IC555.	L3	CO5	05
		OR			
	b.	With a neat diagram explain the concept of summer using Op-amp.	L2	CO1	05
		PART-C			**6X10=60**
8.	a.	What is clipper and explain the types of clippers using OP-amp in detail.	L2	CO1	10
		OR			
	b.	List the characteristics of an ideal op-amp and explain in detail. Give the practical op-amp equivalent circuit.	L2	CO1	10
9.	a.	Describe with relevant waveforms the full wave precision rectifier using Op-amp.	L2	CO2	10
		OR			
	b.	Explain in detail the sample and hold circuit using Op-amp.	L2	CO2	10
10.	a.	Explain in detail the concept of filters using op-amp and its classifications.	L3	CO3	10
		OR			
	b.	Describe in brief how the Op amp is used Schmitt trigger with relevant figures and waveforms.	L3	CO3	10
11.	a.	List and explain the various performance parameters of DAC.	L3	CO4	10
		OR			
	b.	Explain in detail the concept of phase locked loop.	L3	CO4	10
12.	a.	Explain the Structure of a PLA and the way of programming it.	L3	CO5	10
		OR			
	b.	Write a short note on how Digital Logic Gates can be made by connecting together diodes to form different logic structures	L3	CO5	10
13.	a.	Describe in brief the Op amp as integrator and differentiator with relevant figures and waveforms.	L2	CO1	10
		OR			
	b.	With relevant equations and waveforms, explain the Wien bridge oscillator using Op-amp.	L2	CO1	10